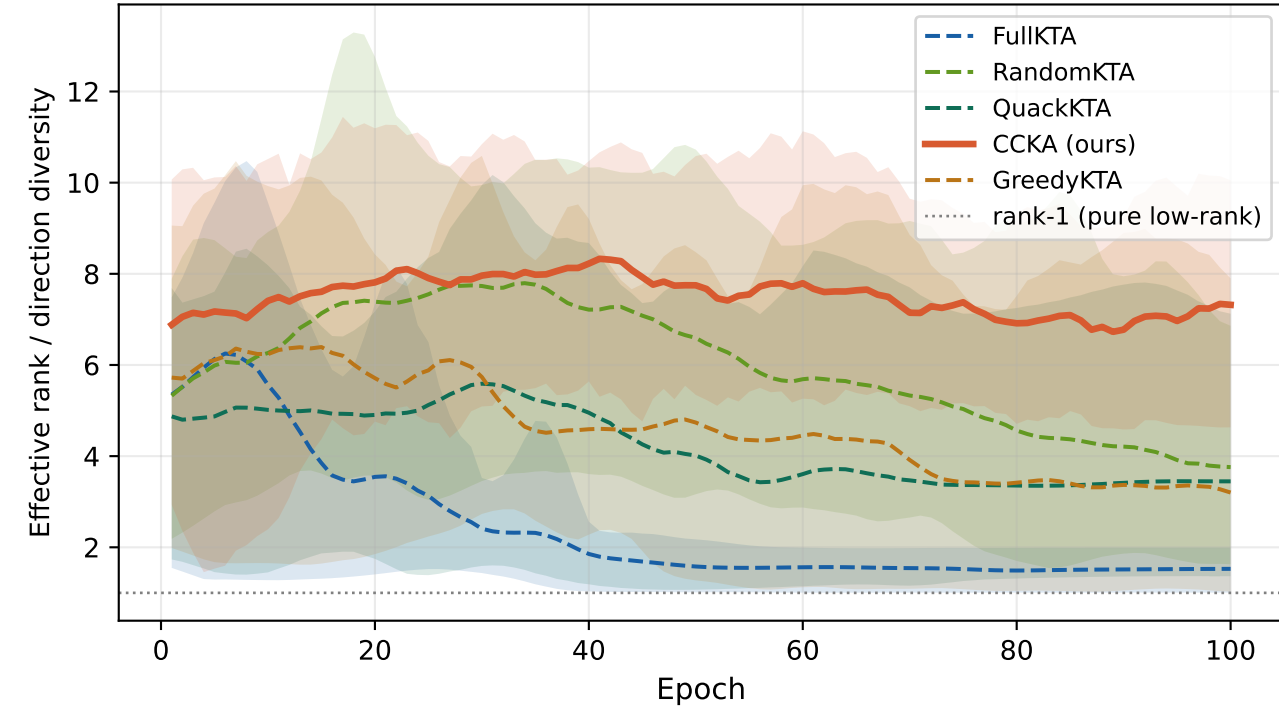


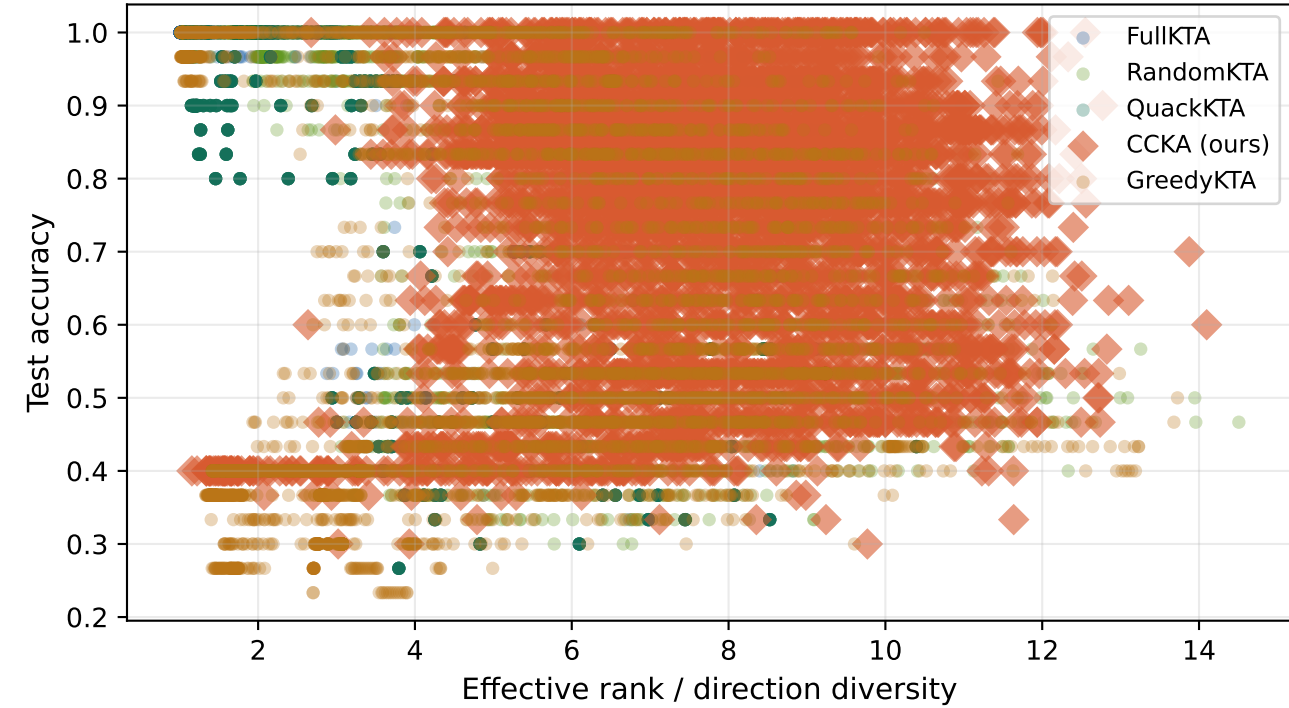
Spectral / Effective-Rank Analysis — CHECKERBOARD (mean $\pm$ min/max, 10 seeds)

Testing the low-rank discriminative alignment theory of CCKA

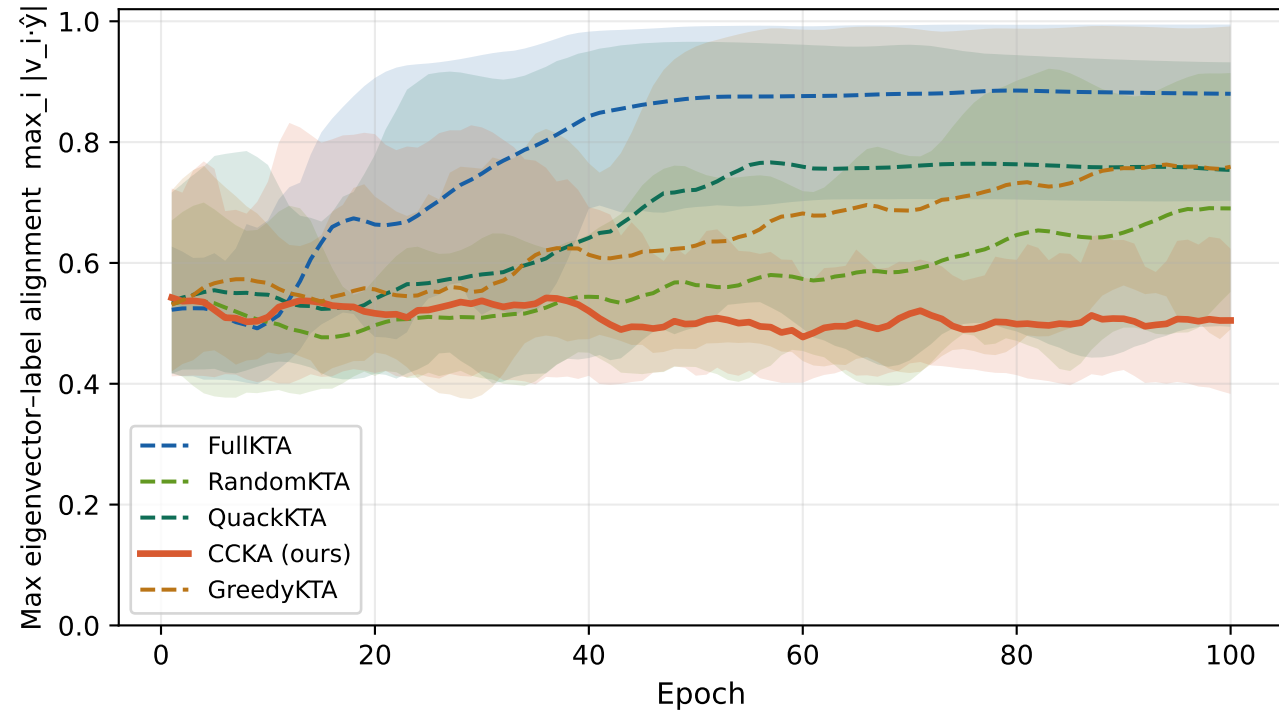
Effective rank (direction diversity) over training — checkerboard
mean $\pm$ min/max, 10 seeds. CCKA expected near 1 (rank-1)



Effective rank vs accuracy — checkerboard (all 10 seeds pooled)
CCKA (◆) achieving high accuracy at LOW eff-rank = rank-1 sufficiency



Best-aligned eigenvector over training — checkerboard
mean $\pm$ min/max, 10 seeds



KTA decomposition by rank — checkerboard (mean $\pm$ min/max, 10 seeds)

